

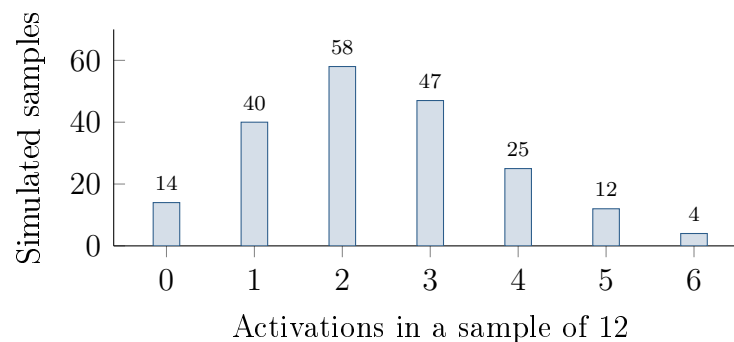
Six Activations in Twelve Users

Unit 3 · Topic 3.3 · THE PROBLEM

Suppose a developer's model says $p_0 = 0.20$ of new app users activate voice commands. An SRS of 12 users includes 6 activations, so $\hat{p} = 0.50$. The chart shows 200 simulated samples of 12 independent users under this model.

Mara: “A normal model for $\hat{p}$ gives $P(\hat{p} \geq 0.50) \approx 0.005$, so that is the chance probability I would report.”

Jules: “The simulated distribution is not very normal. I would estimate the chance probability directly from its tail.”



FIRST TAKE — one sentence before you work the parts

Whose method would you trust more, Mara's smooth curve or Jules's repeated trials? Give one reason from the picture.

- DOK 1** (a) State the most common number of activations in the simulated samples and describe the distribution as roughly symmetric, skewed left, or skewed right.
- DOK 2** (b) Use the chart to estimate $P(\hat{p} \geq 0.50)$ under $p_0 = 0.20$. Then calculate np_0 and $n(1 - p_0)$ and assess the expected-count condition for a normal approximation.
- DOK 3** (c) In 3–4 sentences, choose a method using the simulated tail and either the shape or expected counts. Apply the 5% evidence rule. Explain how reporting 0.005 instead of 0.020 would change a reader's impression.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.